Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Agentic systems engineer with production experience designing, building, and operating multi-agent AI platforms - eval harnesses, guardrails, HITL, RAG pipelines, and agentic process automation. Built **Leasel** - an 8-agent LangGraph cognitive architecture with a reproducible eval harness (225-query benchmark, pass@k multi-trial sampling, LLM-as-judge scoring, regression gating; 93.6% precision) and code-enforced guardrails where LLM reasons and deterministic logic decides every regulated call. Built **AgentProbe** - an agentic testing toolkit with a 5-metric LLM-as-judge CI-integrated eval engine, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), and agent observability dashboard (per-run regression drift, retrieval quality, latency). 8+ years delivering production systems at scale (100K+ MAU, 1.6M financial records, 99.5% uptime). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic AI Architecture | LangChain / LangGraph | Evaluation Harnesses & Offline Evals | Guardrails, HITL & Human-in-the-Loop | RAG Pipelines | Multi-Agent Orchestration | Agent Observability & Monitoring | LLM Cost & Model Routing | MCP | Production AI Systems

WORK EXPERIENCE

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant - Technical Instruction

- Designed and delivered live technical workshops to **600+ students** - Python programming, AI fundamentals, and database design; translated complex technical architectures into hands-on exercises for non-technical audiences.
- One-on-one technical advisory and debugging sessions; rubric-aligned grading and structured feedback across multiple sections.

Magizhs Tech Solution

May 2025 - Aug 2025

AI Solutions Architect - Agentic Systems (Intern)

- **Architected and delivered an end-to-end agentic AI solution** for drone mission planning - defined requirements, designed the cognitive architecture, and deployed to production across marketing, real estate, and commercial survey verticals without a spec doc.
- Solution: LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing), RAG layer over FAA Part 107 regulatory corpus, confidence-scored outputs, validation guardrails enforcing altitude limits and no-fly-zone checks. Self-directed from requirements through production.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate - AI & Data Analytics

- Built multi-source prediction models (Python, R) across 190+ countries - R^2=0.977; applied Isolation Forest anomaly detection surfacing issues manual review missed across 3 research pipelines.

Entrata - PropTech / Real Estate SaaS (100K+ MAU)

Sep 2018 - Aug 2024

Senior Software Engineer

- **Set and enforced technical standards across 20+ production modules** (PHP, Python, React, PostgreSQL, RabbitMQ, AWS) - 300+ peer code reviews driving integration patterns, data models, and compliance guardrails; 99.5% uptime sustained across 100K+ MAU product across 120+ Agile sprints.
- **Shipped Entrata's first production AI pipeline** on 1.6M financial records (Python, PyTorch, OpenAI API) - designed output validation gates, data quality controls, and compliance guardrails from prototype to production with zero incidents.
- **Led microservices redesign** (Docker) - 50% scalability improvement, 30% deployment time reduction; 100% PHI compliance on AWS S3 migration with Libsodium-encrypted pipelines; mentored 5+ junior engineers.
- Optimized 200+ SQL queries (30% performance gain); eliminated 80% of manual operational workload through Python/Bash automation pipelines; defined data standards and KPI definitions for cross-functional stakeholder reporting.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines for 1.6M HUD/Tax Credit records; cloud migration to AWS S3; redesigned document storage database architecture.

PRODUCTION AGENTIC SYSTEMS - REFERENCE IMPLEMENTATIONS

Leasel: 8-Agent AI Platform - Eval Harness, Guardrails, HITL

2026

Designed and built a **production-grade agentic system architecture**: 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → HITL Escalation) with real conditional routing, confidence scoring, and hard safety boundaries enforced in code. **Eval harness**: 225-query benchmark, pass@k multi-trial sampling, LLM-as-judge scoring, regression gating - validates agents in simulation before every release. **Guardrails**: LLM reasons, deterministic logic decides every regulated call; 100% citation correctness; HITL escalation when confidence too low. **Three agentic surfaces**: Document Intelligence (RAG, 91% Precision@1), Support Investigation (automated root-cause analysis, 93.6% precision), and Tenant Self-Service (safety boundary: never auto-resolves eligibility). Audit trail enforced in code: every answer cites exact source clause or is flagged unsupported.

Python · LangGraph · FastAPI · RAG · HITL · Eval Harnesses · LLM-as-Judge · Regression Gating · Guardrails · Audit Trails

AgentProbe: Agentic Testing Toolkit - Multi-Provider LLM Routing & Observability

2024-Present

3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with **multi-provider LLM routing** (Anthropic, OpenAI, Groq, Ollama) - dynamic provider selection for cost/quality/latency tradeoffs. **CI-integrated eval engine**: 5-metric LLM-as-judge scoring (factual accuracy, hallucination, retrieval, completeness, compliance) - regressions blocked before merge. **Agent observability dashboard**: per-run metrics, regression drift detection, retrieval quality, latency monitoring - designed for non-deterministic agents where standard error monitoring is insufficient. 15-endpoint FastAPI backend.

Python · LangGraph · LangChain · FastAPI · ChromaDB · Multi-Provider LLM Routing · LLM-as-Judge · Agent Observability · CI/CD (GitHub Actions)

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University — 2017

SKILLS

Languages: Python, Bash, JavaScript, TypeScript, SQL, PL/SQL, PHP, R

Agentic AI & LLMOps: LangChain, LangGraph, Multi-Agent Orchestration, Agentic AI, Model Context Protocol (MCP), RAG Pipelines, Retrieval-Augmented Generation, Evaluation Harnesses, Offline Evals, Online Evals, Simulation Testing, Regression Gating, LLM-as-Judge, Guardrails, HITL, Human-in-the-Loop, Audit Trails, Policy & Permission Models, Confidence Scoring, Agent Observability, Agent Monitoring, SLO/SLI, LLM Cost Management, Token Management, Model Routing, Prompt Engineering, Tool Calling, FastAPI, ChromaDB, OpenAI API, Anthropic API, Groq, Ollama, spaCy, BERT

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, Distributed Systems, CI/CD (GitHub Actions), PostgreSQL, MySQL, Oracle, RabbitMQ, Git, Linux, Agile/Scrum

Data & ML: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Pandas, NumPy, Tableau, Power BI